# Week42- Assignment-Solution

## Task 1: Writing embedded code

1. Why is memory management more important in writing embedded code?

Because embedded systems are usually constrained by limited storage.

2. Describe the ROM, Flash, and RAM with respect to persistence and the performance for access, and update?

ROM - information stored in the chips is hard-coded at the chips' creation and can only be read afterwards.

Flash - it can store without requiring any power, and so its contents can survive the circuit being unplugged—without the disadvantage of being unchangeable forever more.

RAM - It requires power to retain its contents, but the speed of update is comparable with the time taken to read from it.

3. Mention any two tips for writing efficient code for embedded systems.

Place the more likely to happen code in the if clause.
In some scenarios it is more efficient to insert a value into the code as a plain number rather than to load that value from variable.
Use pointer or a reference to the initial buffer of the Ethernet protocol passed from layer to layer instead copying values.
If required, the system's memory copying and moving routines (such as memcpy and memmove) are usually more efficient.

## Task 2 - Business models

Suppose that a student produces innovative prototype in his/her area of expertise (e.g., intelligent systems) as part of a final year project and thinks to convert it into business. Based on this assumption and as discussed during the lecture, answer the following questions.

1. What are the basic questions that the student must answer to create a business model?

What should we deliver?
Who are the customers?
Who are partners and who are competitors?
How should we sell?
How should we deliver?
How should we get paid?
How should we make a profit?

2. What are the categories of information the student needs in order to develop a business plan?

Knowledge and prerequisites.
Business idea
Products / services

Target groups / market choice
Selected strategies.
Market potential
Organization
The economy of the venture